

1. Suppose  $U$  is an unbiased estimator of  $g(\theta)$ , and  $T$  is sufficient for  $\theta$ . Let  $U^* = \mathbb{E}[U \mid T]$ . Which statement is always true?

A.  $U^*$  has strictly smaller variance than  $U$  for every  $\theta$ .

B.  $U^*$  is the UMVUE of  $g(\theta)$ .

C.  $U^*$  is unbiased for  $g(\theta)$  and has variance no larger than that of  $U$ .

D.  $U^*$  is unbiased only if  $T$  is also complete.

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2. Suppose  $T$  is complete for  $\theta$ , and  $\mathbb{E}_\theta[h(T)] = 5$  for every  $\theta$ . Which conclusion follows?

A.  $h(T) = 0$  with probability one for every  $\theta$ .

B.  $h(T) = 5$  with probability one for every  $\theta$ .

C.  $T$  is sufficient for  $\theta$ .

D. No conclusion follows: completeness applies only to functions with expectation zero.

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3. Let  $X_1, X_2$  be independent  $\mathcal{N}(\theta, 1)$  random variables. Which estimator is the UMVUE of  $\theta^2$ ?

A.  $\frac{1}{2}(X_1^2 + X_2^2) - 1$

B.  $\left(\frac{X_1 + X_2}{2}\right)^2$

C.  $\left(\frac{X_1 + X_2}{2}\right)^2 - \frac{1}{2}$

D.  $X_1 X_2$